KELVIN CHRISTIAN

Data Scientist & Machine Learning Engineer

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EXPERIENCE

Backend & ML Engineer (3-months contract) Feb 2024 – Present
Chekable *Oakland, CA*

- Developed multiple backend APIs for MVP features in a fast-paced agile environment, utilizing Flask, gRPC, and MongoDB
- Participated in the system design discussions to create scalable and high-performance backend architectures
- Orchestrated the seamless deployment and management of 6 microservices on GCP, leveraging Docker and Cloud Run
- Built an automated pipeline to generate patent applications from user's drafts and figures using LangChain and GPT-4o API

NLP Research Intern Jun 2023 – Aug 2023
Institute for Experiential AI *San Jose, CA*

- Worked with 3 researchers to develop a question-answering model for irregular tables using document corpus from ArXiv
- Managed 50 parallel data processing jobs to process 6TB of LaTeX and PDF documents using Slurm, Python, and Bash
- Retrieved 1.7M+ table images and text from PDFs using the Table Transformer and OCR models from Hugging Face
- Implemented a model training pipeline in PyTorch to learn the table representation mapping from table image to LaTeX code

Data Scientist Intern Jul 2022 – Sep 2022
Cisco Systems *Seattle, WA*

- Cleaned and analyzed 1.2TB of real-time DNS queries from AWS S3 using PySpark to enhance the DNS network security
- Reduced ETL time on Databricks by 70% through the optimization of PySpark SQL query and data migration to Delta Lake
- Enhanced user experience by cutting 6% of incorrect website blocks through NLP feature engineering on domain names
- Initiated an anomaly detection system based on users' request patterns by clustering 100,000+ users into 60 profile groups

AI Engineer Intern Jan 2022 – Jun 2022
Aeyesafe *Seattle, WA*

- Developed a microservices-based ETL pipeline on AWS to process real-time sensor data by using RabbitMQ and gRPC
- Utilized Spark and Airflow to automate the generation of weekly business intelligence reports from large-scale datasets
- Wrote 15+ Flask RESTful APIs containing complex SQL queries to manage user profiles, device data, and health metrics
- Led a team of 4 interns in the research and development of diverse machine learning models for fall detection
- Trained and deployed a computer vision model to predict human posture from the thermal camera with 96% test accuracy

PROJECTS

iSEArch Hotels | *Python, Docker, Streamlit, Qdrant, LangChain, OpenAI* Mar 2024

- Built and deployed a RAG hotel recommendation system chatbot by utilizing the Qdrant vector database and GPT-4 model
- Won 2nd place in the Traversaal.ai Hackathon

Diffusion-enhanced Mask Aware Transformer (DeMAT) | *PyTorch, Hugging Face, Tensorboard* Dec 2023

- Developed DeMAT, a deep generative inpainting model that integrates Mask Aware Transformer, Segment Anything, and VAE
- Achieved 11.3% and 39.3% improvements in PSNR and SSIM over Stable Diffusion baseline through extensive experiments

Movie-style Art Generation | *Python, PyTorch, OpenCV, Hugging Face* Dec 2023

- Implemented a pipeline to enhance dataset quality by applying super-resolution on movie frames using Real-ESRGAN model
- Fine-tuned Stable Diffusion model using LoRA and Dreambooth methods, generating high-quality movie-style artwork

EDUCATION

Northeastern University San Jose, CA
M.S. Computer Science, GPA: 3.88/4.00 *2021 – 2023*

University of California, Los Angeles Los Angeles, CA
B.S. Financial Actuarial Mathematics (Minor in Statistics), GPA: 3.58/4.00 *2017 – 2019*

SKILLS

Programming Languages: Python, R, SQL, C++, Java, Javascript
Databases: MySQL, MongoDB, Redis, PostgreSQL, Qdrant, Milvus, Weaviate, Elasticsearch, InfluxDB
Machine Learning: PyTorch, Tensorboard, Spark, OpenCV, NLTK, Scikit-learn, Statsmodels, Pandas, NumPy, Matplotlib, Pyomo
Tools: AWS (EC2, RDS, S3, ECR, Lambda), GCP (GCE, Vertex, Cloud Run), Databricks, Docker, Airflow, OpenAI, LangChain